

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

A truck can haul a maximum weight of **5,630** pounds. During one trip, the truck will be used to haul a **190**-pound piece of equipment as well as several crates. Some of these crates weigh **25** pounds each and the others weigh **62** pounds each. Which inequality represents the possible combinations of the number of **25**-pound crates, x , and the number of **62**-pound crates, y , the truck can haul during one trip if only the piece of equipment and the crates are being hauled?

Answer

- A. $25x + 62y \leq 5,440$
- B. $25x + 62y \geq 5,440$
- C. $62x + 25y \leq 5,630$
- D. $62x + 25y \geq 5,630$

Correct Answer: A

Rationale

Choice A is correct. It's given that a truck can haul a maximum of **5,630** pounds. It's also given that during one trip, the truck will be used to haul a **190**-pound piece of equipment as well as several crates. It follows that the truck can haul at most **5,630 — 190**, or **5,440**, pounds of crates. Since x represents the number of **25**-pound crates, the expression $25x$ represents the weight of the **25**-pound crates. Since y represents the number of **62**-pound crates, $62y$ represents the weight of the **62**-pound crates. Therefore, $25x + 62y$ represents the total weight of the crates the truck can haul. Since the truck can haul at most **5,440** pounds of crates, the total weight of the crates must be less than or equal to **5,440** pounds, or $25x + 62y \leq 5,440$.

Choice B is incorrect. This represents the possible combinations of the number of **25**-pound crates, x , and the number of **62**-pound crates, y , the truck can haul during one trip if it can haul a minimum, not a maximum, of **5,630** pounds.

Choice C is incorrect. This represents the possible combinations of the number of **62**-pound crates, x , and the number of **25**-pound crates, y , the truck can haul during one trip if only crates are being hauled.

Choice D is incorrect. This represents the possible combinations of the number of **62**-pound crates, x , and the number of **25**-pound crates, y , the truck can haul during one trip if it can haul a minimum, not a maximum, weight of **5,630** pounds and only crates are being hauled.